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SYSTEM FOR SELECTIVE INTAKE AIR HEATING

Abstract

A vehicle system, includes an air intake assembly, a heat source, a valve, a temperature indicator and a controller. The air intake assembly including an inlet through which air flows, a filter downstream of the inlet, and an outlet downstream of the filter through which air from the filter flows. The valve is electrically actuated and arranged to selectively permit air flow from the heat source to the inlet. The temperature indicator provides an indication of a temperature in the air intake assembly upstream of the filter. And the controller is communicated with the temperature indicator and the valve and operable to open the valve when the output of the temperature indicator indicates a temperature below a temperature threshold.

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Background/Summary

FIELD

[0001] The present disclosure relates to a system for selectively heating air inducted into an engine air intake.

BACKGROUND

[0002] Vehicles include an air intake through which air is routed to an engine to support combustion in the engine. In cold weather, snow can be taken into or ice may develop within the air intake. The snow or ice can block or otherwise impeded air flow through the air intake assembly and reduce air flow to the engine which negatively impacts engine performance.

SUMMARY

[0003] In at least some implementations, a vehicle system, includes an air intake assembly, a heat source, a valve, a temperature indicator and a controller. The air intake assembly including an inlet through which air flows, a filter downstream of the inlet, and an outlet downstream of the filter through which air from the filter flows. The valve is electrically actuated and arranged to selectively permit air flow from the heat source to the inlet. The temperature indicator provides an indication of a temperature in the air intake assembly upstream of the filter. And the controller is communicated with the temperature indicator and the valve and operable to open the valve when the output of the temperature indicator indicates a temperature below a temperature threshold.

[0004] In at least some implementations, the valve is closed when the temperature in the air intake assembly upstream of the filter is above the temperature threshold.

[0005] In at least some implementations, the heat source is a heat exchanger associated with an engine of the vehicle. In at least some implementations, the heat exchanger is an air-to-liquid heat exchanger through which coolant for an engine is routed.

[0006] In at least some implementations, the heat source is a compressor of a turbocharger.

[0007] In at least some implementations, the heat source is an area outboard of an engine in an engine compartment of the vehicle that is heated by operation of the engine. In at least some implementations, a fan is arranged to move air in a conduit and toward the valve.

[0008] In at least some implementations, the valve has a valve head received in a conduit arranged between the heat source and the inlet, and the valve head is rotatable between a closed position and an open position. In at least some implementations, the valve has an intermediate position between the closed position and the open position. In at least some implementations, a fan provided between the heat source and the valve and arranged to move air in the conduit toward the valve.

[0009] In at least some implementations, the temperature indicator is responsive to a temperature between the inlet and the filter within a housing of the air intake assembly.

[0010] In at least some implementations, a method of selectively increasing the temperature within an air intake for an engine includes determining a temperature; comparing the temperature to a temperature threshold, and providing air flow from a heat source to an intake duct of an air intake assembly when the temperature does not satisfy the temperature threshold.

[0011] In at least some implementations, providing air flow is accomplished by opening a valve to permit air flow through the valve to the intake duct. In at least some implementations, the valve includes a solenoid, and the valve is opened by providing electricity to the valve and the valve is located at least partly in a conduit between the heat source and the intake duct.

[0012] In at least some implementations, the air flow is provided for a predetermined period of time. In at least some implementations, the period of time varies as a function of the magnitude of a difference between the temperature and the temperature threshold.

[0013] In at least some implementations, the temperature is redetermined one or more times and terminating the air flow when the temperature is determined to be above a second temperature which may be equal to or greater than the temperature threshold.

[0014] In at least some implementations, the heat source is a heat exchanger of the vehicle.

[0015] In at least some implementations, the heat source is an engine of the vehicle or a

turbocharger of the vehicle.
[0016] Further areas of applicability of the present disclosure will become apparent from the detailed description, claims and drawings provided hereinafter. It should be understood that the summary and detailed description, including the disclosed embodiments and drawings, are merely exemplary in nature intended for purposes of illustration only and are not intended to limit the scope of the invention, its application or use. Thus, variations that do not depart from the gist of the disclosure are intended to be within the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0017] FIG. **1** is a diagrammatic view of a vehicle including an engine and an air intake system via which air is supplied to the engine;
[0018] FIG. **2** is a diagrammatic view of a system for selectively providing heated air to the air intake assembly;
[0019] FIG. **3** is a schematic view of a system for selectively increasing the temperature within an intake duct of an air intake assembly for a vehicle;
[0020] FIG. **4** is a plan view of a portion of a vehicle engine
[0021] compartment;
[0022] FIG. **5** is a schematic view of a system for selectively increasing the temperature within an intake duct of an air intake assembly for a vehicle;
[0023] FIG. **6** is a plan view of a portion of a vehicle engine compartment; and
[0024] FIG. **7** is a flowchart of a method for selectively increasing the
[0025] temperature within an intake duct of an air intake assembly.
DETAILED DESCRIPTION
[0026] Referring in more detail to the drawings, FIG. **1** illustrates a vehicle **10** having a combustion engine **12** in an engine compartment **14** of the vehicle body **16**. In the engine **12**, a mixture of air and fuel is ignited to generative motive power for the vehicle **10**. Fuel may be provided to the engine **12** by one or more fuel injectors and air may be routed to the engine **12** via an air intake assembly **18**. Engine coolant is circulated relative to the engine **12** and through a heat exchanger or radiator **20**. The radiator **20** is a liquid-to-air heat exchanger by which heat in the coolant is conducted to the metal housing of the radiator **20**, and the heat is radiated and transferred by convection to air surrounding the radiator **20** within the engine compartment **14**. This reduces the temperature of the coolant flowing therethrough and increases the temperature of the surrounding air.
[0027] As shown in FIG. **4**, the air intake assembly **18** may include a housing **22** received in the engine compartment **14** and having an inlet **24** through which air flows into the housing **22**. The inlet **24** may be a port oriented to receive air flowing from the front of the vehicle **10** toward the rear of the vehicle **10**. Air that enters the inlet **24** flows through an intake chamber or duct **26** to a filter **28** located within the housing **22**, downstream of the inlet **24**. The filter **28** removes contaminants from the air flow, and in use, a flow of filtered air exits the filter **28** and is directed by a clean air chamber or clean air duct **30** to an outlet **32** of the air intake assembly **18** from which filtered air exits the housing **22**. Air that exits the housing **22** is routed to the engine **12** to support combustion in the engine **12**.
[0028] In at least some implementations, such as is shown in FIG. **3**, the vehicle **10** includes a turbocharger **34** arranged between the outlet **32** of the air intake assembly **18** and the engine **12**. The turbocharger **34** includes a compressor **36** arranged to discharge air at an increased pressure and provided an increased mass flow rate of air to the engine **12**. The compressor **36** may be coupled to a turbine **37** that is driven, for example, by a flow of exhaust gas from the engine **12**.

[0029] In cold weather, ice can form and snow can enter the inlet **24** of the air intake housing **22**. The ice and snow can restrict air flow to the filter **28** and thus, to the engine **12**, and this can starve the engine **12** of air and decrease engine performance. To inhibit or prevent ice formation, or to melt any ice or snow in the air intake assembly **18**, the vehicle **10** includes a system to provide heated air into the air intake assembly **18**.

[0030] In at least some implementations, as shown in FIG. **2**, the system includes a heat source **38** that gives off heat to air near the heat source **38**, and a conduit **40** communicated with the heat source **38** or air heated thereby, and with the inlet **24** of the air intake housing **22** and via which air may be routed to the interior of the air intake housing **22**, upstream of the filter **28**. In use of the vehicle **10**, air is heated by one or more components of the vehicle, such as heat exchangers (e.g. radiator **20** or other heat exchanger), or the engine **12** or the turbocharger **34**, for example, the compressor **36** thereof, and these components may be considered to be heat sources as can areas near these devices that include the air heated by them.

[0031] In the implementation shown in FIGS. **3** and **4**, compressor **36** and/or area **39** near compressor **36** is a heat source, and heated air may be routed therefrom to the intake duct **26** through the valve **44**, with or without a fan **42** providing a forced air flow. In the implementation shown in FIGS. **5** and **6**, area **41** near engine **12** is a heat source and a fan **42** may be used to force air flow to the intake duct **26** in the conduit **40**. The fan **42** could be a stand alone or dedicated fan, or it may be a fan associated with the radiator **20** or other heat exchanger, or other component in the engine compartment **14**.

[0032] In use, these components or areas often have a temperature above the ambient temperature and heat given off from these components can then be used to increase the temperature within the intake duct **26** of the air intake housing **22**. This temperature increase can inhibit or prevent ice or snow from developing or accumulating within the intake duct **26** of the air intake housing **22** to reduce or eliminate air flow restriction that would otherwise occur from such snow or ice. As used herein, the term “conduit” is intended to mean a component or components that include a void, chamber or passage or other structure(s) by which air may be routed.

[0033] In at least some implementations, a forced air source, like a fan **42**, may provide a forced air flow through the conduit **40** and to the intake duct **26**. In some implementations, with or without a forced air flow from a fan or the like, air may flow through the conduit **40** due to a pressure difference wherein the pressure in the intake duct **26** is lower due to air flow through the intake duct **26** during engine operation or otherwise, or due to a temperature difference causing such air flow.

[0034] In at least some implementations, air through the conduit **40** and to the intake duct **26** of the air intake housing **22** is provided only some of the time and not continually. In such implementations, a valve **44** may be provided to selectively permit and prevent, or at least significantly inhibit, air flow from the conduit **40** and into the air intake housing **22**. The valve **44** may be electrically actuated, such as by including or being driven by a solenoid, to move between a first position and a second position, in which a greater flow or air is permitted than when the valve **44** is in the first position. The valve **44** may be a butterfly valve with a disc-shaped valve head **46** received within the conduit **40**, and in the first position the valve head **46** may mostly or fully block the flow area of the conduit **40** to inhibit or prevent air flow therethrough. In the second position, the valve head **46** is rotated relative to the first position and a greater effective flow area is provided between the valve head **46** and the conduit **40** than when the valve head **46** is in the first position. Flow area, as used herein, is a cross-sectional area through which air may flow within the conduit **40** and around the valve head **46**. In the first position, the valve head **46** may reduce the effective flow area of the conduit **40** by 90% or more, including up to 100%, in at least some implementations. While described with regard to one embodiment as a butterfly valve, the valve **44** may be of any desired construction and shape suitable to control air flow through the conduit **40** as noted herein. In at least some implementations, the valve **44** may also have one or more intermediate positions between the first and second/opened and closed positions, to throttle or otherwise control air flow as desired.

[0035] In at least some implementations, the valve **44** is controlled in response to a temperature, which may be determine by a temperature sensor **48**, and the valve **44** may be opened when the temperature is below a temperature threshold, and the valve **44** may be closed when the temperature is above the temperature threshold. The temperature threshold may be set with regard to a temperature at or below which ice or snow may be likely to form at or near the air intake housing **22**. The temperature may be an ambient temperature (e.g. temperature outside the vehicle **10**), or a temperature at or near the air intake housing **22**, which may include within the intake duct **26** of the air intake housing **22**. In this regard, the temperature sensor **48** may be a sensor used to provide an indication of ambient temperature within the vehicle **10**, such as via an infotainment system of the vehicle **10**, may be obtained from a weather information source (e.g. via a telematics unit of the vehicle **10**), may be a temperature sensor **48** used by other vehicle systems (e.g. to sense the temperature of part of the engine **12** or engine oil temperature) or may be a dedicated temperature sensor used for the purpose of controlling the valve **44**. For example, a temperature sensor **48** may be mounted to the housing **22** and may have a portion exposed to air within the intake duct **26** or otherwise be arranged to be responsive to the temperature within the intake duct **26**. The temperature sensor **48** may provide an output indicative of the temperature within the intake duct **26**, which may be a direct measurement or determination of that temperature or which may be inferred from other temperatures and or time or other information relating to use of the vehicle **10**.

[0036] To control the valve **44**, the system includes a control system **50** with a controller **52** that is communicated with the temperature sensor **48** and with the valve **44**. The controller **52** is operable to open the valve **44** when the output of the temperature sensor **48** indicates a temperature below a temperature threshold, and to close, or permit the valve **44** to move to the closed position such as under the force of a biasing member like a spring, when the temperature is at or above the temperature threshold. In the example wherein the valve **44** includes a solenoid, the controller **52** may selectively provide electricity to the valve **44** to selectively drive the valve **44** and change the position of the valve **44**.

[0037] In at least some implementations, it may be desirable to provide heated air to the intake duct **26** to increase the temperature within the intake duct **26** only in conditions in which snow or ice may exist within the intake duct **26**. In other instances, it might not be desirable to increase the temperature of the air in the intake duct **26** because air at higher temperatures is less dense and leads to reduce engine output. For this reason, the vehicle **10** may include a heat exchanger **56** (e.g. an intercooler or charge air cooler), which may be an air-to-air or liquid-to-air (e.g. liquid coolant or water) heat exchanger, arranged to reduce the temperature of air delivered into the engine **12** for combustion. Thus, in general, it may be desirable to provide colder air to the engine **12** and so, in at least some implementations, heating the intake duct **26** may be limited to conditions in which snow or ice may be present within the intake duct **26**, and to otherwise avoid increasing the temperature within the intake duct **26**. In at least some implementations, the threshold temperature is between 32 degrees Fahrenheit and 36 degrees Fahrenheit.

[0038] In order to perform the functions and desired processing set forth herein, as well as the computations therefore, the control system **50** may include, but is not limited to, one or more controller(s), processor(s), computer(s), DSP(s), memory, storage, register(s), timing, interrupt(s), communication interface(s), and input/output signal interfaces, and the like, referred to be reference numeral **52** in FIG. **2**, as well as combinations comprising at least one of the foregoing. For example, the control system **50** may include input signal processing and filtering to enable accurate sampling and conversion or acquisitions of such signals from communications interfaces and sensors. As used herein the terms control system **50** may refer to one or more processing circuits such as an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality. The control system **50** may be distributed among different vehicle modules, such as an infotainment control module, engine control module or unit, powertrain control module, transmission control module, and the like, if desired.

[0039] The term “memory” or “storage” as used herein can include volatile memory and/or non-volatile memory, generally referred to by reference numeral **54** in FIG. **2**. Non-volatile memory can include, for example, ROM (read only memory), PROM (programmable read only memory), EPROM (erasable PROM), and EEPROM (electrically erasable PROM). Volatile memory can include, for example, RAM (random access memory), synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), and direct RAM bus RAM (DRRAM). The memory can store an operating system that controls or allocates resources of a computing device.

[0040] FIG. **7** illustrates a method **60** of controlling temperature within an air intake assembly **18** of a vehicle **10**. The method **60** may be run after the vehicle **10** is turned on (e.g. keyed on, or engine started, or the like) and in step **62**, a temperature is determined. The temperature may be determined from information from any suitable temperature indicator, which may be a source of temperature data within or remotely located from the vehicle **10**, or a temperature sensor **48** of the vehicle **10**, for example.

[0041] Next, in step **64**, the determined temperature is compared to a threshold temperature. If the determined temperature is not less than the threshold temperature, then the method **60** ensure that the valve **44** is in the second position, inhibiting or preventing air flow from the heated air source to the air intake assembly **18**, and if not, the method **60** may skip to step **66** and cause the valve **44** to move to the second position (or remain in the second position if the valve is already in the second position) and then the method **60** may end. If a dedicated fan **42** is provided to cause a forced air flow to the intake duct **26**, then this step may also include turning off the fan or ensuring that the fan remains off, so that the fan is used only when the valve **44** is open.

[0042] If desired, instead of ending after one iteration, the method **60** may loop back to step **62** one or more times to check or determine the temperature again and compare again the newly determined temperature to the threshold temperature in step **64**. A counter may be used and incremented in step **68** each time the determined temperature is not less than the threshold temperature and the method **60** may end after the counter reaches a desired number as checked in step **70**. In at least some implementations, the counter value needed to terminate the method **60** may be set as a function of the difference between the determined temperature and the threshold temperature with a greater differential requiring fewer iterations (down to one iteration), as the confidence that snow or ice conditions are not present is greater when the determined temperature is higher.

[0043] If in step **64** the determined temperature is less than the threshold temperature, then the method **60** continues to step **72** in which it is determined if the valve **44** is in the first position in which greater air flow is permitted by the valve **44**. If the valve **44** is not in the first position, then the method **60** proceeds to step **74** in which the valve **44** is moved to the first position (e.g. opened) to permit air flow, or an increased air flow, from the heated air source to the air intake assembly **18**. If a dedicated fan **42** is used, this step may also include turning on the fan **42** to provide a forced air flow so that the fan is operated only when the valve **44** is open, if desired. In the example wherein the temperature determination is based on the output from a temperature sensor **48** that is responsive to the instantaneous or current temperature within the air intake assembly **18** (e.g. within the intake duct **26**) then the method **60** may loop back to step **62**, perhaps after expiration of a time period determined by a timer associated with the controller **52** to allow some time for the temperature in the air intake assembly **18** to increase or with use of a counter as noted above. This new, updated or more current temperature determination is then compared to the threshold temperature in step **64** or to a second temperature which may be a set value or a variable threshold, and which may be the same as or different than the original or first temperature threshold.

[0044] In other example methods, the system may assume a temperature increase in the air intake assembly **18** occurs after a certain amount of time after the engine **12** has been operating, and after this time, the method **60** may move the valve **44** to the second position and then end. This assumption of temperature increase may be made based on empirical data taken over time and related to the temperature within the engine compartment **14** of the vehicle **10** as a function of operation of the engine **12**, engine oil temperature or other engine temperature, or the like, which may be mapped or otherwise determined as a function of or independently of an ambient temperature. The time period may be varied based upon the difference between the determined temperature and the threshold temperature, with a greater time period provided when the determined temperature is lower.

[0045] For example, if the determined temperature is ten degrees Fahrenheit below the threshold temperature, then the method **60** may cause the valve **44** to be opened (in the first position) for a predetermined time period deemed sufficient to ensure that the temperature within the intake duct **26** increases at least ten degrees before the valve **44** is closed (moved to the second position). In this example, when the determined temperature is five degrees Fahrenheit below the threshold temperature, the time period might be less. In this way, further iterations of redetermining the temperature and comparing the redetermined temperature to the threshold temperature are not needed. This may facilitate use of ambient temperature such as may be determined by a temperature sensor **48** of the vehicle **10** or a remote temperature/weather data source, or an engine temperature, to conveniently be used to determine when the valve **44** should be opened so that a dedicated temperature sensor is not needed for the air intake assembly **18** in some implementations.

[0046] The systems and method herein can, among other things, prevent or inhibit or reduce ice or snow within the intake duct of the air intake assembly to improve air flow through the assembly and to the engine. The system may intelligently control air flow to the air intake assembly to provide heated air when needed and avoid unduly heating air when heated air is not needed within the air intake assembly.

Claims

1. A vehicle system, comprising: an air intake assembly including an inlet through which air flows, a filter downstream of the inlet, and an outlet downstream of the filter through which air from the filter flows; a heat source; a valve that is electrically actuated and arranged to selectively permit air flow from the heat source to the inlet; a temperature indicator providing an indication of a temperature in the air intake assembly upstream of the filter; and a controller communicated with the temperature indicator and the valve and operable to open the valve when the output of the temperature indicator indicates a temperature below a temperature threshold.

2. The system of claim 1 wherein the valve is closed when the temperature in the air intake assembly upstream of the filter is above the temperature threshold.

3. The system of claim 1 wherein the heat source is a heat exchanger associated with an engine of the vehicle.

4. The system of claim 3 wherein the heat exchanger is an air-to-liquid heat exchanger through which coolant for an engine is routed.

5. The system of claim 1 wherein the heat source is a compressor of a turbocharger.

6. The system of claim 1 wherein the heat source is an area outboard of an engine in an engine compartment of the vehicle that is heated by operation of the engine.

7. The system of claim 6 which also includes a fan arranged to move air in a conduit and toward the valve.

8. The system of claim 1 wherein the valve has a valve head received in a conduit arranged between the heat source and the inlet, and the valve head is rotatable between a closed position and an open position.

9. The system of claim 8 wherein the valve has an intermediate position between the closed position and the open position.

10. The system of claim 8 which also includes a fan provided between the heat source and the valve and arranged to move air in the conduit toward the valve.

11. The system of claim 1 wherein the temperature indicator is responsive to a temperature between the inlet and the filter within a housing of the air intake assembly.

12. A method of selectively increasing the temperature within an air intake for an engine, comprising: determining a temperature; comparing the temperature to a temperature threshold; providing air flow from a heat source to an intake duct of an air intake assembly by opening a valve to permit air flow through the valve to the intake duct when the temperature does not satisfy the temperature threshold; and preventing air flow from the heat source to the intake duct of the intake assembly by closing a valve to prevent air flow through the valve to the intake duct when the temperature satisfies the temperature threshold.

13. The method of claim 12 wherein providing air flow is accomplished by opening a valve to permit air flow through the valve to the intake duct.

14. The method of claim 13 wherein the valve includes a solenoid, and the valve is opened by providing electricity to the valve and the valve is located at least partly in a conduit between the heat source and the intake duct.

15. The method of claim 12 wherein the air flow is provided for a predetermined period of time.

16. The method of claim 15 wherein the period of time varies as a function of the magnitude of a difference between the temperature and the temperature threshold.

17. The method of claim 12 which also includes redetermining the temperature one or more times and terminating the air flow when the temperature is determined to be above a second temperature which may be equal to or greater than the temperature threshold.

18. The method of claim 12 wherein the heat source is a heat exchanger of the vehicle.

19. The method of claim 12 wherein the heat source is an engine of the vehicle or a turbocharger of the vehicle.
